

# Lead Management System

Implementation plan, clickable-level screen prototype and day-to-day operating guide for the **Owner Console** → **Lead** module (gmtraders-lead-microservice).

- 
- |   |                              |    |                                       |
|---|------------------------------|----|---------------------------------------|
| 1 | Goals & scope                | 7  | How to manage leads (operating guide) |
| 2 | How a lead moves (lifecycle) | 8  | Roles & permissions                   |
| 3 | Modules & features           | 9  | API plan                              |
| 4 | System architecture          | 10 | Delivery roadmap                      |
| 5 | Data model                   | 11 | KPIs, risks & next steps              |
| 6 | Prototype – screens          |    |                                       |

# 1 Goals & scope

Every customer enquiry — WhatsApp, Facebook, Instagram, website, phone or walk-in — lands in one place, gets an owner within minutes, is followed up on time, and ends as either a **customer** or a **recorded loss with a reason**.

## 1 · Capture everything

Leads arrive automatically from Meta channels (WhatsApp Cloud API, Messenger, Instagram DM) and the website enquiry form. Staff add phone-call and walk-in leads by hand in under 30 seconds.

## 2 · Never miss a follow-up

Each open lead always has a next follow-up date. Overdue follow-ups are shown on the Overview and to the assigned person first thing each day.

## 3 · Know what sells

Reports show which source brings paying customers, conversion rate per staff member, pipeline value by stage, and why deals are lost.

## In scope (Owner Console → Lead menu)

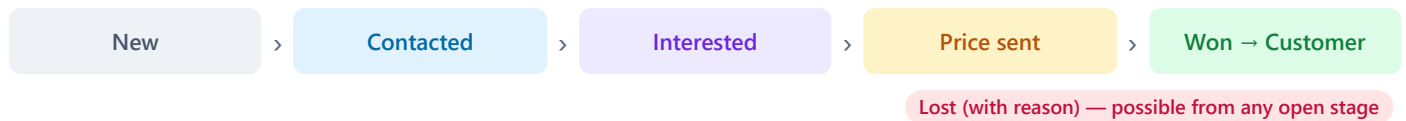
| Menu          | What it does                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Overview      | KPIs, today's follow-ups, pipeline by stage, leads by source, staff leaderboard.                                               |
| Leads         | List/search/filter, create, edit, assign, mark hot, change stage, add notes, timeline, "Make customer".                        |
| Messages      | Unified inbox for WhatsApp / Facebook / Instagram with 24-hour reply window, templates, "Save as lead".                        |
| Sales stages  | Kanban board (New → Contacted → Interested → Price sent → Won / Lost), drag to move.                                           |
| Follow-ups    | Today / overdue / upcoming lists, complete with result, reschedule.                                                            |
| Reports       | Source performance, conversion funnel, staff performance, lost-reason analysis, export to Excel.                               |
| Lead sources  | Connect channels (Meta accounts, website form key), activate/deactivate sources.                                               |
| Bulk messages | WhatsApp broadcasts using approved templates to an audience (hot, by stage, by source, no-reply-in-N-days) with cost estimate. |
| Auto-replies  | Rules: new lead greeting, outside office hours, keyword match, no reply after N hours; auto-assign & mark hot.                 |
| Settings      | Stages, lost reasons, assignment rules (fixed user / round-robin), message templates, web enquiry forms.                       |

## Out of scope (for now)

- Quotation / invoice creation — handled by the existing Sale service; the lead only stores *price sent* amount and links to the customer after conversion.
- TikTok, LinkedIn, YouTube, Email channel integrations — kept as manual sources until APIs are justified by volume.
- Voice call recording / telephony integration.

## 2 How a lead moves (lifecycle)

The six default stages are created per business by Setup/EnsureDefaults. Businesses can rename, recolour, reorder or add stages; every stage has a type — **Open**, **Won** or **Lost**.



### End-to-end flow

| #  | Step            | What happens                                                                                                                                                                   | Automatic?                      |
|----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1  | Enquiry arrives | Meta webhook or website form hits the service. A <b>Conversation</b> is created/updated and the <b>Message</b> stored (duplicate ExternalMessageId skipped).                   | Yes                             |
| 2  | Auto-reply      | Matching <b>AutoReplyRule</b> fires: greeting, office-hours reply, keyword reply (e.g. "price", "rate", "dam").                                                                | Yes                             |
| 3  | Save as lead    | Website/phone leads are created directly. Chat contacts become a <b>Lead</b> via "Save as lead" (or automatically on first message if the rule says so). LeadNo = LD-00001...  | Web form: yes · Chat: one click |
| 4  | Assign          | <b>LeadAssignmentRule</b> by priority: fixed user or round-robin across a team, optionally <i>mark hot</i> . Activity "Assigned" logged.                                       | Yes (manual override)           |
| 5  | First contact   | Salesperson calls or replies, moves lead to <b>Contacted</b> , sets next follow-up.                                                                                            | Manual                          |
| 6  | Qualify         | Requirement (e.g. "5,000 boxes, 3-colour print") and expected value captured → <b>Interested</b> .                                                                             | Manual                          |
| 7  | Price sent      | Quote sent; activity <i>PriceSent</i> with amount → <b>Price sent</b> .                                                                                                        | Manual                          |
| 8  | Follow-up loop  | <b>LeadFollowUp</b> reminders until the customer decides. "No reply after N hours" rule can nudge automatically.                                                               | Reminders automatic             |
| 9a | Won             | "Make customer" publishes an event to the Sale service (RabbitMQ), receives CustomerId, sets WonDate. Activity <i>Converted</i> .                                              | One click                       |
| 9b | Lost            | Lost reason is mandatory (Too costly, Order too small, No reply, Bought elsewhere, Delivery too long, Other). Sets LostDate. Lead can be re-targeted with bulk messages later. | Manual                          |

**Rule:** an Open lead must always have a NextFollowUpDate. The API rejects a stage change to an Open stage without one; this single rule is what keeps the pipeline clean.

## 3 Modules & features

Grouped by priority. P1 = must-have for go-live, P2 = soon after, P3 = growth features.

### Leads core P1

- Create / edit / soft-delete lead; duplicate check on phone
- Filters: stage, source, assigned user, hot, date range, overdue
- Activity timeline (note, call, message, price sent, stage change, assigned, converted)
- Change stage with validation (lost reason, next follow-up)
- Assign / reassign, mark hot

### Follow-ups P1

- Types: call, message, visit, email
- Today / overdue / upcoming tabs, "My" vs "All"
- Complete with result → auto-create next follow-up
- Daily reminder notification to assigned user

### Setup masters P1

- Stages (name, colour, order, type)
- Lead sources, lost reasons
- Defaults seeded per business (done)

### Overview & reports P1

- KPIs: new leads, open pipeline %, won %, conversion %, overdue
- Source / staff / funnel / lost-reason reports, Excel export

### Make customer P2

- RabbitMQ event to Sale service → CustomerId back
- Lead is read-only once converted (except notes)

### Website enquiry form P2

- Public endpoint secured by FormKey + allowed domain
- Copy-paste HTML/JS snippet, thank-you message, rate-limit

### Messages inbox P2

- Meta webhooks (WhatsApp, Messenger, Instagram)
- 24-hour reply window countdown; template required after
- Unread counts, assign conversation, save as lead

### Assignment rules & auto-replies P2

- Fixed user / round-robin by source, priority order
- Triggers: new lead, outside hours, keyword, no reply after N hours

### Templates & bulk messages P3

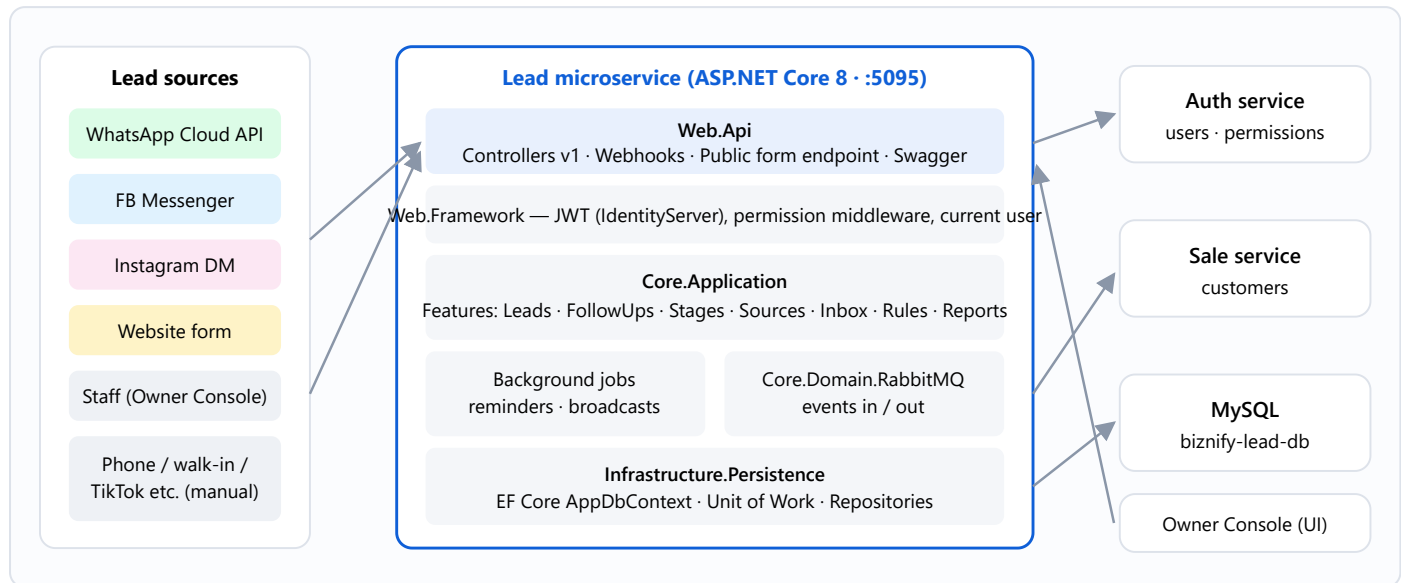
- Template approval sync with Meta (Utility vs Marketing price)
- Audience builder, schedule, cost estimate, delivery/reply stats

### Quality & ops P3

- Audit trail, encrypted access tokens, webhook signature check
- Background jobs: reminders, scheduled broadcasts, no-reply rule

## 4 System architecture

Built on the same template as the other Biznify services. Clean layers, CQRS-style features (Command/Query + Handler), permission-checked controllers.

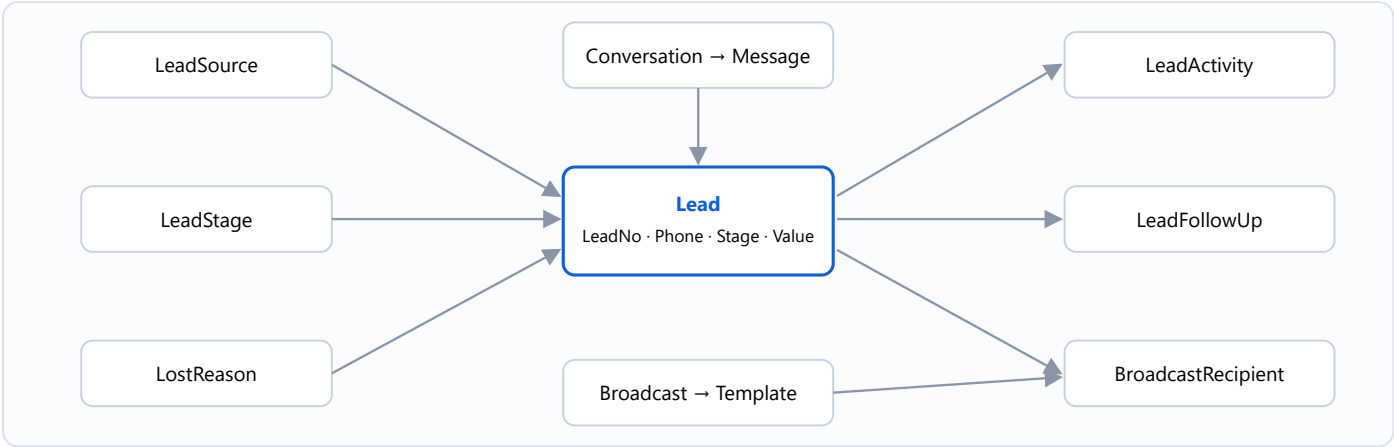


### Key technical decisions

| Topic               | Decision                                                                                                                         |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Multi-business      | Every row carries the business id from BaseEntity; all queries filter by the current user's business.                            |
| Staff               | Users live in the Auth service; the lead DB stores only AssignedUserId (string). Names are resolved and cached per request.      |
| Webhooks            | Verify Meta X-Hub-Signature-256; respond 200 fast, process via queue; idempotent on ExternalMessageId.                           |
| Secrets             | LeadSource.AccessToken encrypted at rest (Data Protection); never returned by the API.                                           |
| Customer conversion | Async event <i>LeadConvertToCustomer</i> → Sale service creates/links customer → reply event sets CustomerId.                    |
| Scheduling          | Hosted background service (or Hangfire) for follow-up reminders, scheduled broadcasts and the no-reply rule — every 1–5 minutes. |
| Deploy              | Publish only Web.Api; Staging = GM demo DB, Production = GM live DB.                                                             |

# 5 Data model

14 entities and 15 enums already exist in the InitialLead migration (Step 0). The core relationships:



| Entity                                           | Purpose / key fields                                                                                                                                                                            |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lead                                             | LeadNo, Name, CompanyName, Phone, Email, SocialHandle, Source, Stage, Requirement, ExpectedValue, AssignedUserId, IsHot, NextFollowUpDate, LostReason, Won/LostDate, CustomerId, ConversationId |
| LeadActivity                                     | Timeline row: type (Note, StageChange, Call, Message, PriceSent, Assigned, FollowUp, Converted), old/new stage, amount, user (empty = automatic)                                                |
| LeadFollowUp                                     | Type (Call/Message/Visit/Email), DueDate, AssignedUserId, Note, IsDone, DoneDate, Result                                                                                                        |
| LeadStage / LeadSource / LostReason              | Per-business masters. Stage has SortOrder, Color, StageType (Open/Won/Lost). Source has SourceType, connection info, encrypted token.                                                           |
| Conversation / Message                           | Chat thread per external contact & channel; unread count, 24-h reply window; messages with direction, type, status, auto-reply rule id.                                                         |
| LeadAssignmentRule                               | Source filter, FixedUser or RoundRobin (UserIds, LastAssignedUserId), MarkAsHot, Priority                                                                                                       |
| AutoReplyRule                                    | Trigger (NewLead, OutsideOfficeHours, KeywordMatch, NoReplyAfterHours, NewLeadFromSource), keywords, hours, action (SendMessage, AssignUser, MarkHot, NotifyStaff)                              |
| MessageTemplate / Broadcast / BroadcastRecipient | Meta-approved templates (Utility/Marketing), audience filter, schedule, delivery counters and cost estimate, per-recipient status                                                               |
| WebEnquiryForm                                   | FormKey, AllowedDomain, DefaultAssignedUserId, ThankYouMessage                                                                                                                                  |

**Suggested indexes:** (BusinessId, LeadStageId), (BusinessId, AssignedUserId, NextFollowUpDate), (BusinessId, Phone), Message.ExternalMessageId unique, Conversation (LeadSourceId, ExternalContactId) unique.

## 6 Prototype – screens

Wireframe-level prototype of the Owner Console → Lead menu. Sample data is illustrative (print-business enquiries).

GM Owner Console

Overview

Leads

Messages7

Sales stages

Follow-ups

Reports

Lead sources

Bulk messages

Auto-replies

Settings

Overview

This month ▾

+ New lead

New leads

184▲12%

Open pipeline

₹ 18.4 L

Won this month

₹ 6.2 L▲8%

Conversion

21%

Overdue follow-ups

9

Pipeline by stage

|            |             |    |
|------------|-------------|----|
| New        | <div></div> | 46 |
| Contacted  | <div></div> | 38 |
| Interested | <div></div> | 25 |
| Price sent | <div></div> | 17 |
| Won        | <div></div> | 39 |
| Lost       | <div></div> | 19 |

Today's follow-ups

☐ Call · Rahim Uddin

10:30

☐ Msg · Nabila Fashion

11:00

☒ Visit · Karim Pharma

09:00

☐ Call · Sultana Traders **overdue**

Yest.

Leads by source

|          |     |          |     |
|----------|-----|----------|-----|
| WhatsApp | 41% | Facebook | 27% |
| Website  | 14% | Walk-in  | 9%  |

Screen 1 • Overview. The first screen every morning: KPIs, stage funnel, and the user's own follow-ups for today (overdue in red).

GM Owner Console

Overview

Leads

Messages

Sales stages

Follow-ups

Reports

Lead sources

Bulk messages

Auto-replies

Settings

Leads (184)

Export

+ New lead

Search name / phone / LD-no

Stage: All ▾

Source: All ▾

Assigned: Me ▾

Hot only

Overdue

| Lead     | Name / Company                        | Phone        | Source       | Stage      | Value      | Assigned | Next follow-up          |
|----------|---------------------------------------|--------------|--------------|------------|------------|----------|-------------------------|
| LD-00184 | <b>Rahim Uddin</b><br>Rahim Sweets    | 01711-xxxxxx | WhatsApp     | Price sent | ₹ 45,000   | Sajib    | Today 10:30             |
| LD-00183 | <b>Nabila Akter</b><br>Nabila Fashion | 01819-xxxxxx | Instagram    | Interested | ₹ 22,500   | Mitu     | Today 11:00             |
| LD-00182 | <b>Karim Hossain</b><br>Karim Pharma  | 01915-xxxxxx | Website form | Contacted  | ₹ 1,20,000 | Sajib    | 30 Sep                  |
| LD-00181 | <b>Tanjila Rahman</b><br>xxxxxx       | 01612-xxxxxx | Facebook     | New        | —          | Mitu     | Today 15:00             |
| LD-00179 | <b>Sultana Traders</b><br>xxxxxx      | 01552-xxxxxx | Phone call   | Contacted  | ₹ 18,000   | Rafi     | <b>27 Sep (overdue)</b> |
| LD-00176 | <b>Arif Printing Hub</b><br>xxxxxx    | 01756-xxxxxx | Walk-in      | Won        | ₹ 64,000   | Rafi     | —                       |

Screen 2 • Leads list. Default filter is “Assigned: Me”. Owner/manager switches to All. Row click opens the lead detail.

GM Owner Console

Overview

Leads

Messages

Sales stages

Follow-ups

Reports

Lead sources

Bulk messages

Auto-replies

Settings

LD-00184 · Rahim Uddin 🔥 Hot

Open chat

Mark lost

Make customer

New ✓

>

Contacted ✓

>

Interested ✓

>

Price sent

>

Won

Activity timeline

+ Note

Log call

Price sent

+ Follow-up

- **Price sent** · ₹ 45,000 — 2,000 sweet boxes, 4-colour, laminated. Sajib · Today 09:12
- **Stage** Interested → Price sent Sajib · Today 09:12
- **Call** — wants delivery before 10 Oct; asked for sample. Sajib · Yesterday
- **Assigned** to Sajib by rule "WhatsApp round-robin" Automatic · 25 Sep
- **Auto-reply** "Thanks for contacting GM Print..." Automatic · 25 Sep
- **Lead created** from WhatsApp chat 25 Sep 21:40

Details

Company

Rahim Sweets, Mirpur-10

Phone / WhatsApp

01711-xxxxxx

Source

WhatsApp

Requirement

Sweet boxes 2,000 pcs, 4-colour offset + lamination

Expected value

₹ 45,000

Assigned to

Sajib ▾

Next follow-up

**Today 10:30 · Call** — confirm sample approval

**Screen 3 · Lead detail.** Stage stepper, full timeline (manual + automatic events), one-click actions. "Mark lost" opens a reason picker; "Make customer" links to the Sale service.

console.gmprint.com / lead / sales-stages

Sales stages

Assigned: All ▾ Source: All ▾

| New                               | 46 | Contacted                          | 38 | Interested                        | 25 | Price sent                         | 17 | Won                           | 39 | Lost                       | 19 |
|-----------------------------------|----|------------------------------------|----|-----------------------------------|----|------------------------------------|----|-------------------------------|----|----------------------------|----|
| Tanjila Rahman<br>Facebook · Mitu |    | Karim Pharma<br>₹ 1,20,000 · Sajib |    | Nabila Fashion<br>₹ 22,500 · Mitu |    | Rahim Sweets 🔥<br>₹ 45,000 · Sajib |    | Arif Printing Hub<br>₹ 64,000 |    | Mou Boutique<br>Too costly |    |
| Hasan Motors<br>Website · Rafi    |    | Sultana Traders<br>overdue · Rafi  |    | Green School<br>₹ 80,000 · Rafi   |    | City Hospital<br>₹ 2,10,000 · Rafi |    | Delta Foods<br>₹ 1,05,000     |    | Rupali Events<br>No reply  |    |
| Joy Bakery<br>WhatsApp · Sajib    |    |                                    |    |                                   |    |                                    |    |                               |    |                            |    |

**Screen 4 · Sales stages board.** Drag a card to change stage; dropping on Lost asks for a reason, dropping on an Open stage asks for the next follow-up date. Column totals show count and ₹ value.

console.gmprint.com / lead / messages

Messages

All channels ▾UnreadMine

Rahim Uddin WA

Sample ta kobe pabo?

Nabila Akter IG 2

Price koto hobe 500 pcs?

Hasan Motors FB

Banner size 10x4 ft...

+880 1612... WA 1

Visiting card rate?

Rahim Uddin · WhatsApp

Reply window: 18h 12m left

Assalamu alaikum, sweet box print korte chai 2000 pcs.

🤖 Thanks for contacting GM Print Solution! Our team will reply within 15 minutes.

Walaikum salam. 4-colour + lamination e ₳ 45,000 porbe, delivery 7 din.

Sample ta kobe pabo?

Type a reply... /templates

Send

Contact

Lead

LD-00184 · Price sent

Assigned

Sajib ▾

Quick actions

Open lead

+ Follow-up

Mark hot

Unknown contacts show a **Save as lead** button here.

**Screen 5 · Messages.** One inbox for WhatsApp, Facebook and Instagram. Meta allows free-form replies only within 24 hours of the customer's last message; after that a WhatsApp template must be used.

/ lead / follow-ups

Follow-ups

+ Add

Overdue 9

Today 14

Upcoming 31

Done

| Lead                                     | Type    | Due           |
|------------------------------------------|---------|---------------|
| <input type="checkbox"/> Sultana Traders | Call    | <b>27 Sep</b> |
| <input type="checkbox"/> Rahim Sweets    | Call    | 10:30         |
| <input type="checkbox"/> Nabila Fashion  | Message | 11:00         |
| <input type="checkbox"/> Tanjila Rahman  | Call    | 15:00         |
| <input type="checkbox"/> City Hospital   | Visit   | 16:30         |

Ticking "done" asks for the result and the next follow-up in the same dialog.

/ lead / auto-replies

Auto-replies

+ Rule

| Rule         | Trigger                   | Action       | On                                  |
|--------------|---------------------------|--------------|-------------------------------------|
| Welcome      | New lead                  | Send message | <input checked="" type="checkbox"/> |
| After hours  | Outside 10:00–20:00       | Send message | <input checked="" type="checkbox"/> |
| Price query  | Keyword: price, rate, dam | Mark hot     | <input checked="" type="checkbox"/> |
| Nudge        | No reply after 24 h       | Notify staff | <input checked="" type="checkbox"/> |
| Website lead | New lead from Website     | Assign Rafi  | <input type="checkbox"/>            |

Assignment rules (Settings) decide *who* owns a new lead; auto-replies decide *what the customer hears*.

**Screens 6–7 · Follow-ups and Auto-replies.**

/ lead / bulk-messages / new

New bulk message

Schedule

1 · Template (Meta approved)

"Eid offer – 15% off packaging"

Approved

Marketing

2 · Channel

WhatsApp – 01xxx GM Print

3 · Audience

Lost – Too costly

No reply 30 days

4 · When

05 Oct 2026, 11:00

Recipients

312

Estimated cost

₹ 1,590

Opted-out removed

14

/ lead / reports

Reports · Source performance

Excel

| Source    | Leads | Won | Conv. | Won ₹ |
|-----------|-------|-----|-------|-------|
| WhatsApp  | 76    | 19  | 25%   | 2.9 L |
| Facebook  | 50    | 7   | 14%   | 0.9 L |
| Website   | 26    | 8   | 31%   | 1.6 L |
| Walk-in   | 17    | 4   | 24%   | 0.6 L |
| Instagram | 15    | 1   | 7%    | 0.2 L |

Lost reasons

Too costly

8

No reply

5

Delivery too long

3

**Screens 8–9 · Bulk messages and Reports.** Other report tabs: Conversion funnel, Staff performance, Follow-up discipline (on-time %), Response time.

/ lead / lead-sources

Lead sources

+ Source

| Source                           | Type     | Status        |          |
|----------------------------------|----------|---------------|----------|
| WhatsApp                         | Meta API | Connected     | Manage   |
| Facebook                         | Meta API | Connected     | Manage   |
| Instagram                        | Meta API | Not connected | Connect  |
| Website form                     | Form key | Active        | Get code |
| Phone call · Walk-in · TikTok... | Manual   | Active        | —        |

/ lead / settings

Settings

StagesLost reasonsAssignmentTemplatesWeb forms

| ↕ | Stage      | Type | Colour |
|---|------------|------|--------|
| ⋮ | New        | Open | slate  |
| ⋮ | Contacted  | Open | sky    |
| ⋮ | Interested | Open | violet |
| ⋮ | Price sent | Open | amber  |
| ⋮ | Won        | Won  | green  |
| ⋮ | Lost       | Lost | rose   |

**Screens 10–11 · Lead sources and Settings.** A stage that has leads cannot be deleted — only renamed or merged.

New-lead dialog (manual entry)

Name \*

Phone \* (duplicate check)

Company

Source \*

Requirement

Expected value ₹

Assign to (blank = rule)

First follow-up \*

Cancel

Save lead

## 7 How to manage leads (operating guide)

The system only works if the team follows a simple rhythm. This is the recommended routine for GM Traders.

### One-time setup (Owner / Admin — about 1 hour)

| # | Task                                                                                                                  | Where                         |
|---|-----------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 1 | Run <b>Set up defaults</b> — creates stages, sources and lost reasons.                                                | Settings (auto on first open) |
| 2 | Adjust stage names/colours if needed (e.g. add “Sample sent” between Interested and Price sent).                      | Settings → Stages             |
| 3 | Give staff the Lead permissions (see section 8).                                                                      | Auth → Roles                  |
| 4 | Connect WhatsApp Business number and Facebook/Instagram page.                                                         | Lead sources                  |
| 5 | Create the website enquiry form and paste the snippet on the website.                                                 | Settings → Web forms          |
| 6 | Create assignment rules: e.g. WhatsApp + Facebook → round-robin Sajib/Mitu; Website → Rafi; price keyword → mark hot. | Settings → Assignment         |
| 7 | Turn on Welcome and After-hours auto-replies (Bangla + English).                                                      | Auto-replies                  |
| 8 | Submit WhatsApp templates for approval: follow-up reminder (Utility), offer (Marketing).                              | Settings → Templates          |

### Daily routine

#### Morning (10 min)

- Open **Overview** → check overdue follow-ups first.
- Open **Follow-ups** → **Today** and work top-down.
- Check **Messages** → **Unread**; reply within 15 min during office hours.

#### During the day

- Every call/message: log it on the lead (one click).
- Move the stage the moment it changes.
- After every contact, set the **next follow-up**.
- Walk-in or phone enquiry → + **New lead** immediately.

#### End of day (5 min)

- Follow-ups → Today should be empty (done or rescheduled).
- No lead in “New” older than 24 h.
- Hot leads: confirm each has a follow-up tomorrow.

### Weekly (Owner / Sales manager — 30 min, e.g. Saturday)

- **Sales stages board**: look for cards stuck in the same stage for more than 7 days — reassign or close as lost.
- **Reports** → **Staff performance**: leads handled, conversion %, follow-up on-time %, first response time.
- **Reports** → **Source performance**: shift ad budget toward sources with the best *won*  $\hat{z}$ , not just the most leads.
- **Lost reasons**: if “Too costly” dominates, review pricing or offer a smaller-quantity option.
- Plan one **bulk message** campaign for lost/no-reply leads (only with approved templates, respect opt-outs).

### Golden rules for the team

1. **Every enquiry is a lead** — no enquiry lives only in someone's phone.
2. **Every open lead has an owner and a next follow-up date.**
3. **Log it or it didn't happen** — calls, prices and promises go on the timeline.

4. **Lost needs a reason** — this is how we learn.
5. **Reply fast** — first response under 15 minutes in office hours.
6. **Won** → **Make customer** the same day so Sales/Accounts see it.

## 8 Roles & permissions

27 permission codes are already defined in `migrations-sql/lead-permissions.sql`. Suggested role bundles:

| Permission group                | Codes                                                               | Owner | Manager          | Sales staff       |
|---------------------------------|---------------------------------------------------------------------|-------|------------------|-------------------|
| Setup                           | leadSetup                                                           | ✓     | —                | —                 |
| Stages / sources / lost reasons | leadStage*, leadSource*, lostReason*<br>(List/Create/Update/Delete) | ✓     | List +<br>Update | List              |
| Leads                           | leadList, leadDetails, leadCreate, leadUpdate                       | ✓     | ✓                | ✓ (own)           |
| Lead delete                     | leadDelete                                                          | ✓     | ✓                | —                 |
| Stage / assign / note           | leadChangeStage, leadAssign, leadNoteCreate                         | ✓     | ✓                | Stage +<br>Note   |
| Follow-ups                      | leadFollowUpList, leadFollowUpCreate,<br>leadFollowUpComplete       | ✓     | ✓                | ✓                 |
| Overview & reports              | leadDashboard, leadReport                                           | ✓     | ✓                | Overview<br>(own) |
| Convert                         | leadConvertToCustomer                                               | ✓     | ✓                | ✓                 |

**To add before the messaging phases** (not yet in the SQL file): `leadInboxView`, `leadMessageSend`, `leadConversationAssign`, `leadTemplateList/Create/Update/Delete`, `leadBroadcastList/Create/Send`, `leadAutoReplyList/Create/Update/Delete`, `leadAssignmentRuleList/Create/Update/Delete`, `leadWebFormList/Create/Update/Delete`, `leadSourceConnect`, `leadViewAll` (see all users' leads, not just own).

### Data visibility

- **Sales staff** see leads, follow-ups and conversations assigned to them (plus unassigned conversations if they have `leadInboxView`).
- **Manager / Owner** with `leadViewAll` see everything in the business and can reassign.
- Reports respect the same rule — staff see only their own numbers.

## 9 API plan

All routes under /api/v1, JWT-protected and permission-checked, except the webhook and public form endpoints.

| Controller                                                        | Endpoints                                                                                                                                                                                                |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Setup ✓                                                           | POST Setup/EnsureDefaults                                                                                                                                                                                |
| LeadStage / LeadSource / LostReason                               | GET list · GET {id} · POST · PUT {id} · DELETE {id} · PUT reorder                                                                                                                                        |
| Lead                                                              | GET (paged, filters) · GET {id} · POST · PUT {id} · DELETE {id} · PUT {id}/stage · PUT {id}/assign · PUT {id}/hot · POST {id}/note · POST {id}/activity · POST {id}/convert · GET check-duplicate?phone= |
| LeadFollowUp                                                      | GET (today/overdue/upcoming/done) · POST · PUT {id} · PUT {id}/complete                                                                                                                                  |
| Dashboard / Report                                                | GET overview · GET pipeline · GET by-source · GET by-staff · GET funnel · GET lost-reasons · GET export                                                                                                  |
| Conversation / Message                                            | GET conversations · GET {id}/messages · POST {id}/messages · PUT {id}/assign · POST {id}/save-as-lead · PUT {id}/read                                                                                    |
| Webhook (public)                                                  | GET/POST webhook/meta (verify token + signature)                                                                                                                                                         |
| WebEnquiry (public)                                               | POST public/enquiry/{formKey} (domain + rate limit)                                                                                                                                                      |
| AssignmentRule / AutoReplyRule / MessageTemplate / WebEnquiryForm | CRUD + PUT {id}/toggle · Template: POST {id}/submit-to-meta · POST sync-status                                                                                                                           |
| Broadcast                                                         | GET · POST (draft) · POST {id}/estimate · POST {id}/schedule · POST {id}/send · GET {id}/recipients                                                                                                      |

## 10 Delivery roadmap

Built in small, releasable steps. Each step ships API + Owner Console screens + permissions, tested on Staging (GM demo) before Production (GM live). Estimates assume 1 backend + 1 frontend developer.

| Step                                                                                                                        | Deliverable               | Scope                                                                                                                                                          | Est. | Status |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|
| 0                                                                                                                           | Foundation                | Project scaffold, 14 entities + 15 enums, InitialLead migration, EnsureDefaults, 27 permission codes, environment config                                       | —    | Done ✓ |
| 1                                                                                                                           | Setup masters             | CRUD for stages (reorder, colour, type), sources, lost reasons; Settings screens                                                                               | 3 d  | Next   |
| 2                                                                                                                           | Leads core                | Lead CRUD, LeadNo generator, duplicate check, filters/paging, stage change rules, assign, hot, notes, activity timeline; Leads list + detail + new-lead dialog | 6 d  |        |
| 3                                                                                                                           | Follow-ups                | Follow-up CRUD, complete-with-next, NextFollowUpDate sync, daily reminder job; Follow-ups screen                                                               | 3 d  |        |
| 4                                                                                                                           | Board & overview          | Kanban (drag = stage change), Overview KPIs                                                                                                                    | 3 d  |        |
| 5                                                                                                                           | Reports                   | Source, staff, funnel, lost-reason reports; Excel export                                                                                                       | 3 d  |        |
| 6                                                                                                                           | Make customer             | RabbitMQ contract with Sale service, CustomerId round-trip, read-only after conversion                                                                         | 2 d  |        |
| Milestone A — Go-live for manual leads (≈ 4 weeks). Team starts using it for phone, walk-in and copy-pasted chat enquiries. |                           |                                                                                                                                                                |      |        |
| 7                                                                                                                           | Website form              | Web form CRUD, public endpoint, embed snippet, anti-spam                                                                                                       | 2 d  |        |
| 8                                                                                                                           | Assignment rules          | Fixed / round-robin by source & priority, mark hot                                                                                                             | 2 d  |        |
| 9                                                                                                                           | Meta channels & inbox     | Meta app review, webhook, WhatsApp/Messenger/Instagram send & receive, conversations, 24-h window, save as lead; Messages screen                               | 8 d  |        |
| 10                                                                                                                          | Auto-replies              | Rule engine for 5 triggers × 4 actions, office hours, no-reply job                                                                                             | 3 d  |        |
| Milestone B — Omnichannel (≈ +3.5 weeks). All chat enquiries flow in automatically.                                         |                           |                                                                                                                                                                |      |        |
| 11                                                                                                                          | Templates & bulk messages | Template approval sync, audience builder, cost estimate, scheduled sending, delivery/reply webhooks                                                            | 5 d  |        |
| 12                                                                                                                          | Hardening                 | Load test, audit log, token encryption review, monitoring, user training & handover                                                                            | 3 d  |        |

**Long-lead item:** Meta Business verification and WhatsApp app review can take 1–3 weeks. Start it during Step 1 so it is ready by Step 9.

### Definition of done (every step)

- Endpoints documented in Swagger, permission attribute on every action, business-id filter verified.
- Validation messages in SetupResources-style resource files; errors via the standard response wrapper.
- Migration generated and applied to demo DB; tested on Staging by the GM team with real sample data.
- Owner Console screen matches the prototype; works on a 1366px laptop and on mobile width.

## 11 KPIs, risks & next steps

### How we measure success (after 60 days)

| KPI                                | Target              | Where to see it                           |
|------------------------------------|---------------------|-------------------------------------------|
| Enquiries captured as leads        | 100%                | Messages without a lead vs. leads created |
| First response time (office hours) | < 15 min            | Reports → Response time                   |
| Open leads with a next follow-up   | 100%                | Overview                                  |
| Follow-ups done on time            | > 85%               | Reports → Follow-up discipline            |
| Lead → customer conversion         | +5 pts vs. baseline | Reports → Funnel                          |
| Lost leads with a reason           | 100%                | Reports → Lost reasons                    |

### Risks & mitigations

| Risk                                               | Mitigation                                                                                      |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Meta approval delays / policy changes              | Start verification early; Milestone A does not depend on Meta; keep manual sources working.     |
| Staff don't log activities                         | One-click actions, mandatory follow-up on stage change, weekly review of the discipline report. |
| Duplicate leads (same phone on WhatsApp & walk-in) | Normalise phone (+880), duplicate check on create, merge action later.                          |
| WhatsApp cost overrun on broadcasts                | Mandatory cost estimate + owner confirmation, Utility vs Marketing categories, opt-out list.    |
| DB not reachable from outside the server           | Development via SSH tunnel (documented in README); deploy only Web.Api.                         |
| Access-token leaks                                 | Encrypt at rest, never return in API, rotate on disconnect.                                     |

### Immediate next steps

1. Review this plan and prototype with the GM owner; confirm stage names and the lead fields.
2. Start Meta Business verification for the WhatsApp number and Facebook/Instagram page.
3. Build **Step 1 — Setup masters** (API + Settings screens).
4. Add the extra permission codes listed in section 8 to `lead-permissions.sql` before Step 7.
5. Agree the RabbitMQ contract with the Sale service team for "Make customer".